Agentic AI with Structured Skill Files

Autonomously Generates GPU-Accelerated Virtual Screening

FPR2 (CHEMBL4227) · PDB 7T6S · Claude Code Opus 4.6

01 THE CONCEPT

Claude Code (LLM Agent) + **Skill File (212 lines)** → **Rigorous VS Pipeline**

Domain knowledge injection via plain-text — no fine-tuning, no RAG

02 AUTONOMOUS PIPELINE

1	**ChEMBL API Query**	382 FPR2 actives (pChEMBL ≥ 5)
2	**Butina Clustering**	100 diverse actives (TC 0.4, ECFP4)
3	**Decoy Generation**	4,780 property-matched decoys (MW/LogP/HBD/HBA/RotB)
4	**Structural Filtering**	PAINS A/B/C + Brenk filters (762 compounds removed)
5	**Ligand Preparation**	RDKit ETKDGv3 + Meeko 0.7.1 → PDBQT
6	**GPU Docking**	Uni-Dock 1.1.3 · 2× RTX 4500 Ada · ~5,000 ligands

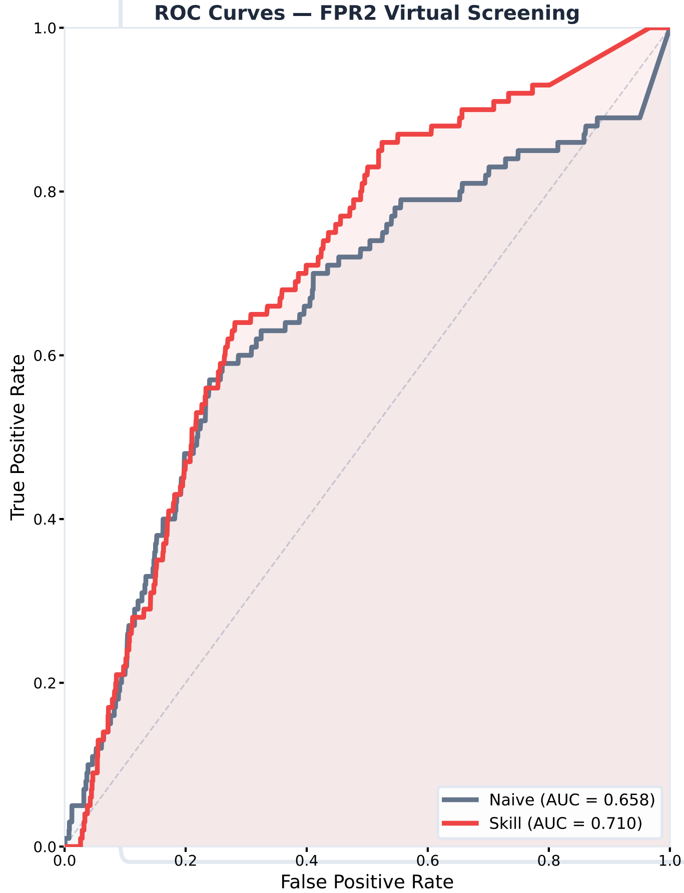
18 scripts | **~3,100 lines** | **zero human editing**

03 PROTOCOL COMPARISON

Parameter	Naive	Skill-Guided
Ligand 3D	OpenBabel gen3d	**RDKit ETKDGv3**
PDBQT Prep	OpenBabel	**Meeko 0.7.1**
Receptor Prep	OpenBabel	**Meeko mk_prepare_receptor**
Struct. Filters	None	**PAINS A/B/C + Brenk**
Box Size	20 Å	**25 Å**
Exhaustiveness	8	**32**

04 RESULTS

ROC Curves — FPR2 Virtual Screening



ROC AUC Improvement

+0.048

DeLong p = 0.004

Naive	Skill
0.658	**0.710**

$EF_{10\%}$: 2.2 (skill) vs 2.1 (naive)
Mann–Whitney U: $p < 10^{-10}$ (both)

Naive: 4,632 docked (89 actives, 4,543 decoys)
Skill: 4,082 docked (93 actives, 3,989 decoys)

95% CI: [0.591, 0.713] naive
95% CI: [0.650, 0.748] skill

Lightweight skill files democratise rigorous large-scale virtual screening